

Product Manual

Kamoer Fluid Tech (Shanghai) Co.,Ltd.

Version: A/1

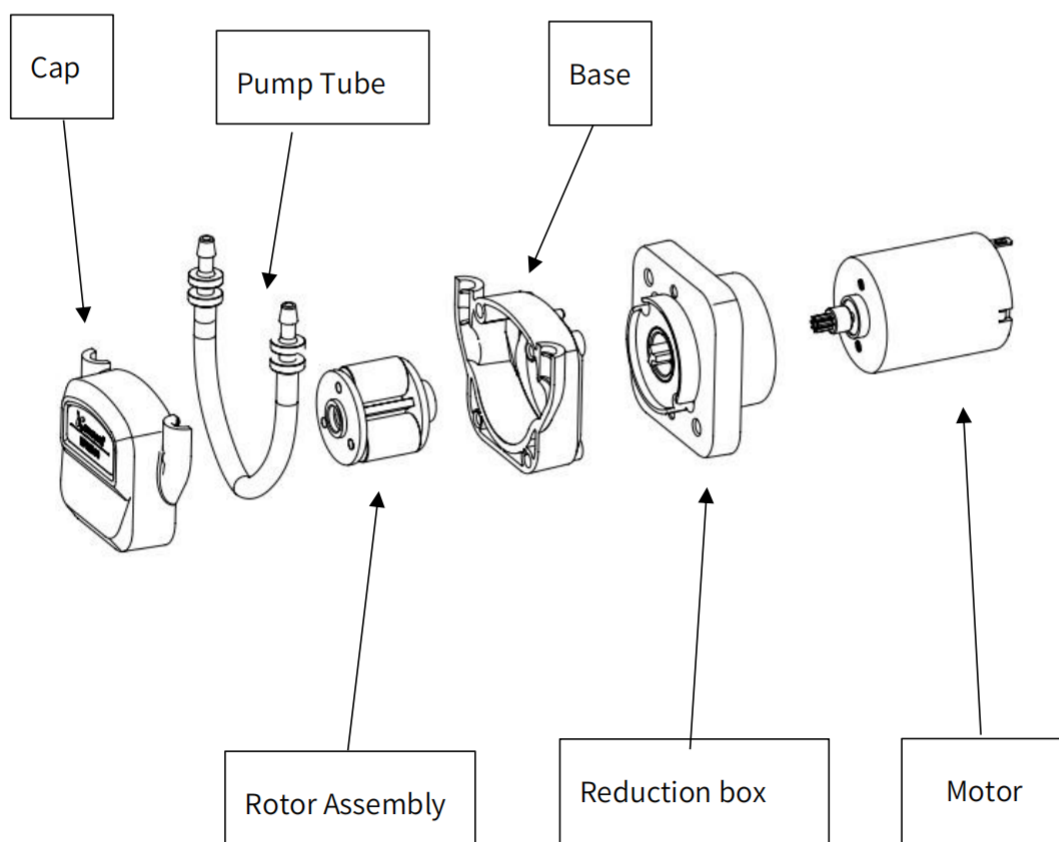
Product name	Micro peristaltic pump
Model	KPHM100
Release Date	2022.11.23
Manufacturer	Kamoer Fluid Tech (Shanghai) Co.,Ltd.

1. Product Overview

1.1 Product Images



1.2 Product Structure



1.3 Model Explanation

KPHM100	HA	Null	B	10	Null
Model	Motor Type	Reduction	Tube Material	Tube Size	Mounting

1.3.1 Motor Type



HA: 24V DC brushed motor

HB: 12V DC brushed motor



HD: 24V DC brushless motor

HE: 24V DC brushless motor



ST: 24V stepper motor

1.3.2 Deceleration

1: 1-stage deceleration, reduction ratio 1:14 0: No deceleration (stepper motor only)

1.3.3 Pump Tube Material

B: BPT tube

1.3.4 Pump Tube Size

10#: 3mm×5mm

1.3.5 Mounting

Blank: standard installation; L: with L type bracket

1.3.6 Pump Material

Pump Head	Pump Cap	Base	Rotor	Rotor Assembly	Tube fittings
KPHM100	PC plastic	POM	PET	POM	PP



1.4 Features & Applications

1.4.1 Features

- KPHM100 series peristaltic pumps are 3 types of motor driven peristaltic pumps, 24V/12V motor, different voltage options are available
- Pump tube selection - the pump tube adopts imported BPT (B) tube
- This peristaltic pump can be adapted to common brushed DC, brushless DC and stepper motor
- Ultra-small size, compact structure and light weight
- Small installation method - the installation method is fixed through the plate, in addition, the pump head adopts the quick release type, and the orientation of the pipe joint can be selected in any direction of up, down, left and right
- Transmission mode - The transmission mode is gear transmission, the first-stage reduction ratio is 1:14, with high reliability and low noise

1.4.2 Applications

Analytical instruments: ion chromatograph, blood analyzer, hypochlorous acid water generator;
detergent delivery: automatic car washer.



Ion chromatograph



Auto car washer



Blood analyzer



Hypochlorous Acid

1.5 Patents & Certifications

CN306503568S



RoHS certificate



CE certificate



- ◆ The tube is resistant to liquid media. It is necessary to check chemical compatibility or perform immersion experiments. Improper tube selection may cause the tube to be damaged quickly.
- ◆ The working environment of the product should not exceed 45°C, and the humidity should not exceed 70% (no condensation). The harsh working environment will cause premature damage to the product.



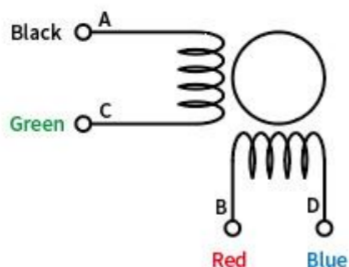
- ## Make it smart to pump fluid





2.3.3 Stepper Motor Parameters & Wiring

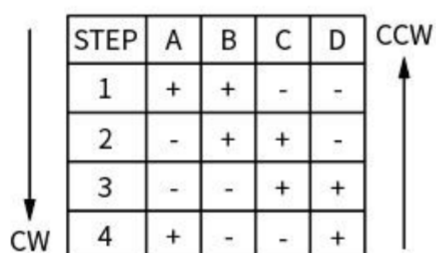
Wiring diagram:

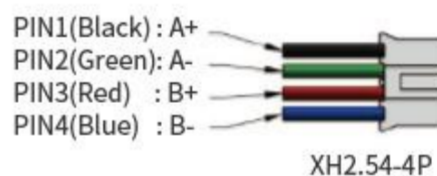


Pin No. VS. Lead Wire Color

PIN NO.	Color
1	Black
2	Green
3	Red
4	Blue

Exciting Sequence vs. Direction of Rotation

STEP	A	B	C	D	
1	+	+	-	-	
2	-	+	+	-	
3	-	-	+	+	
4	+	-	-	+	

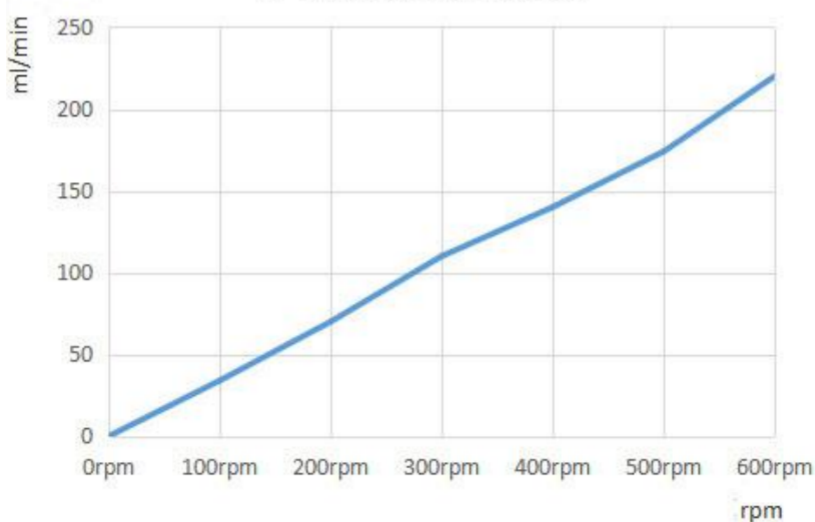


Clockwise View from Motor Mounting Side

Phase	Resistance	Inductance	Current	Holding torque	Insulation class
2	$2.5 \pm 10\% \Omega$	$2.5 \pm 20\% \text{mH}$	1.0A	0.18Nm	B

2.3.4 Flow Rate Curve

KPHM100 Flow Curve



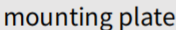
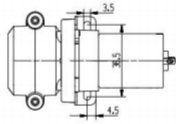
Flow curve is for reference only, and the specific parameters are subject to the actual working condition test.



3. Pump Tube

Tube Material	Features	Standards compliant	Tube Code
BPT tube (B) 	Good resistance to common chemicals, acid and alkali, and oxidation resistance, life time >1000h.	RoHS, FDA	10#

4. Optional accessories

Accessory name	Specifications		Function
L shape mounting plate 	56.6×31.2×13mm		Increase installation methods to meet different customer needs
Straight connector	1/8"		External connection can be connected to 3 × 5 mm hard pipe or similar hose
Stepper motor driver	MODBUS-RTU		It can realize built-in and external potentiometer speed regulation, can realize MODBUS RS485 communication and 0-10V analog speed regulation, realize electric and trigger two start and stop modes, the maximum subdivision is 32, and the drive current can be set (0.7A, 1.2A), 1.7A, 2.2A, 2.7A, 2.9A, 3.2A, 4.0A)
Pulse generator board + Stepper driver board	2802 + KMD-42B-P		Support clockwise and counterclockwise operation, LED 4-digit digital tube display, various control methods: rotary encoder, foot switch, external analog, 485 communication control

Special declaration: The company reserves the right to improve the appearance and technical specifications of the product without prior notice.

